

R5 — ML Research Track

Member: _____

Your role

You run the ple arm (seed 1). You own the quantization (PTQ) verification for the whole project.

Weekly tasks

Week 1 — PTQ explainer

- Task: Read `research/tinystories/quantize_eval.py` and `src/quantize.py`. Write the explainer: group-wise symmetric int4 PTQ, group 64 vs 128, fp16 scales.
- Deliverable: The 1-page PTQ explainer.
- Verify: R2 can answer a quiz from the explainer.
- Learn: Neural network basics. Watch the 3Blue1Brown series on neural networks.

Week 2 — Launch ple seed 1

- Task: Launch the run: `uv run python -m research.tinystories.train --arm ple --vocab 32768 --steps 3000 --target-core 559000 --seed 1` on your Colab account.
- Deliverable: The run reaches 3000 steps. The checkpoint and the run JSON record land in `runs/`.
- Verify: Checkpoints save to Google Drive every 1000 steps.
- Learn: Embeddings and next-token prediction. Explain why the ple table is 25M parameters.

Week 3 — PTQ group 64

- Task: Run the group-64 PTQ eval for both seeds: `uv run python -m research.tinystories.quantize_eval --tag cleandeploy --seed 0` and `--seed 1`.
- Deliverable: The PTQ degradation table (fp32 vs 4-bit) for all arms.
- Verify: Your table matches the published shape: ple degrades less than baseline.
- Learn: The transformer block. Explain where the weights of one block live.

Week 4 — PTQ report

- Task: Write the PTQ report. Does the ple gain survive 4-bit in our runs? The published claim is 124-128% retained.
- Deliverable: The PTQ section of the verification report.
- Verify: Your numbers sit near the published retention range.
- Learn: Perplexity and evaluation. Explain why a big redundant table survives quantization.

Week 5 — Eval protocol design

- Task: Design the eval protocol for the improvise model: ppl plus a human coherence score.
- Deliverable: The protocol doc.
- Verify: Two members score the same text and land within 1 point.
- Learn: RoPE and KV cache. Draw the cache growth over 200 tokens.

Week 6 — Quantize the domain model

- Task: Quantize the fine-tuned model with PTQ. Verify the quality stays.
- Deliverable: The 4-bit domain model and its ppl.
- Verify: The ppl delta from fp32 stays near the published delta.

- Learn: Fine-tuning and PTQ. Explain the order: fine-tune first, then quantize.

Week 7 — Teach LayerNorm

- Task: Present LayerNorm and residual connections to the team at the meetup.
- Deliverable: Your teach-back slides.
- Verify: H4 can explain why residuals help training.
- Learn: Prepare your presentation deeply.

Week 8 — What I learned

- Task: Write the "what I learned" page. One page, honest.
- Deliverable: MY_LEARNING.md in the club repo.
- Verify: You can teach the page to a new member in 10 minutes.
- Learn: Your own writing is the learning.

Learning path

1. Neural network basics: perceptron, MLP, gradient descent.
2. Embeddings, tokenization, next-token prediction.
3. Transformer block: attention, LayerNorm, residuals, MLP.
4. Loss, perplexity, sampling, generation.
5. Attention math, RoPE, KV cache. PLE paper.
6. Fine-tuning, transfer learning, PTQ.
7. Teach one topic to the team.
8. Write the "what I learned" page.

Resources

- Repo: <https://github.com/slvDev/esp32-ai>
- RESULTS.md quantization section: <https://github.com/slvDev/esp32-ai/blob/main/RESULTS.md>
- 3Blue1Brown neural networks: https://www.youtube.com/playlist?list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi